

Chapter 3: The Internet and Electronic Mail

The Internet is a global collection of computer networks, often called the "Information Super Highway." It enables the world to function as a "Global Village" by facilitating rapid information sharing.

3.1 Internet Fundamentals

- **URL (Uniform Resource Locator):** The unique address of a web resource.
Example: `http://www.google.com`.
- **Protocol:** A set of rules for data transmission. Common examples include **HTTP** (Hypertext Transfer Protocol) and **FTP** (File Transfer Protocol).
- **IP Address:** A unique numerical address (e.g., 173.252.110.27) used to identify a computer on the Internet.
- **Domain Name:** A user-friendly name (e.g., google.com) that represents the IP address.
- **DNS (Domain Name Server):** A service that converts domain names into their corresponding IP addresses.

3.2 Web Browsers and Search Engines

Web Browsers: Software used to access and view websites (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge).

Search Engines: Websites designed to help users find information using keywords (e.g., Google, Yahoo, Ask.com).

3.3 Electronic Mail (E-mail)

E-mail is a service for exchanging digital messages. An e-mail address consists of two parts separated by the @ symbol:

[Username] @ [Domain Name]

- **Advantages:** Low cost, high speed, ability to send attachments, and accessibility from anywhere.
- **Common Folders:** Inbox (received mail), Sent (sent mail), Drafts (unsaved mail), and Spam (junk mail).

3.4 Modern Communication Services

- **Video Conferencing:** Conducting live, visual connections between two or more people in different locations (e.g., Skype, Zoom).
- **Social Networking:** Platforms used to build social relations and share content (e.g., Facebook, LinkedIn, Twitter).
- **Cloud Computing:** Using remote servers on the Internet to store, manage, and process data instead of a local computer.

3.5 Ethics and Safety

The **Internet Society** is a non-profit organization that helps maintain the ethics and principles of Internet usage. Users should be aware of security threats like viruses and use tools like anti-virus software and firewalls.